NOTES 14: RANDOMIZATION TESTS AND P-VALUES

Stat 120 | Spring 2026

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Example: Resume callbacks. Researchers want to know if unconscious gender bias affects hiring decisions. They created 20 identical, highly qualified resumes. On 10 of the resumes, they put a traditionally male name (like "John"). On the other 10, they put a traditionally female name (like "Emily"). They submitted all 20 resumes to open job postings and waited to see who received a callback for an interview. Of the 20 resumes, 10 received callbacks. 7 of those callbacks went to "John" and 3 went to "Emily". Is this evidence of gender bias?

- Population parameter:
- Sample statistic:
- Hypotheses:

How do we gauge whether we have strong evidence against the null hypothesis, or if the difference could be due to sampling variability?

Procedure:

- 10 cards labeled "Yes" (those that received a callback)
- 10 cards labeled "No" (did not receive a callback)

1. Verify you have 10 yes's and 10 no's
2. Shuffle the 20 cards together thoroughly
3. Deal the cards face down into two equal piles of 10 (Pile 1 is the "Emily" group, Pile 2 is the "John" group)
4. Flip over the "John" group and count the number of yes's
5. Record your result and then pass the cards to the next person

Your group should perform the shuffle and deal 4 times (1 per person). Record the number of "Yes" cards in the Male Name pile for each trial.

Trial Number	Number of Callbacks ("Yes" Cards) in the "John" Pile	Sample Statistic
Trial 1		
Trial 2		
Trial 3		
Trial 4		

StatKey Example: Resume callbacks

Constructing null distribution for a difference in proportions

Combine all cases, then randomly assign each case to one of two groups with the same sample sizes as the original data. Record the difference in proportions between the two groups. Do this many times.

StatKey Example In a random sample of 765 adults in the United States, 322 say they could not cover a $400 unexpected expense without borrowing money or going into debt. A journalist claims that the true proportion is actually 50%. Is the journalist justified?

Constructing null distribution for a single proportion

In this example, flip a coin 765 times and record the proportion that came up heads. Do this many times to construct the randomization distribution

R Example In a study about Universal Basic Income, the SEED institute randomly selected 100 residents and gave them an unrestricted $500 a month for 24 months (the "UBI" group). At the end of the study period, they were given an emotional well-being assessment and results were compared to 100 control-group residents who did not receive payments. The average well-being score among the UBI group was 31.5 and among the control group was 29.5. Does UBI result in higher well-being scores?

```r
ubi |>
  group_by(group) |>
  summarize(
    mean = mean(well_being),
    n = n()
  )
```

```
# A tibble: 2 x 3
  group     mean     n
  <chr>    <dbl> <int>
1 Control   29.5   100
2 UBI       31.5   100
```

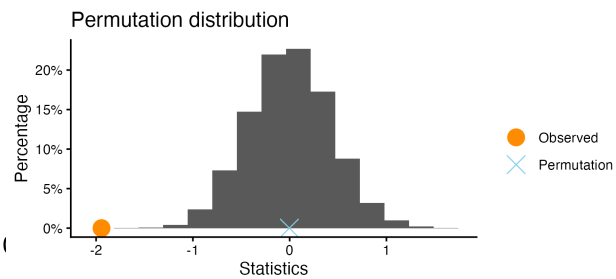
```r
library(CarletonStats)
permTest(well_being ~ group, data = ubi, alternative = "greater", seed = 101525)
```

```
    ** Permutation test **

 Permutation test with alternative: greater
 Observed statistic
  Control :  29.54   UBI :  31.49
 Observed difference: -1.944

 Mean of permutation distribution: -0.00282
 Standard error of permutation distribution: (
 P-value:  1

    *-------------*
```



> **Constructing null distribution for a difference means**
>
> Combine all cases, then randomly assign each case to one of two groups with the same sample sizes as the original data. Record the difference in means between the two groups. Do this many times.

Be careful about direction of hypotheses!

Statistical Significance

Significance Level

Cutoff value for rejecting H_0 (called "alpha" or α)
If p-value _______ α, we "reject H_0" and say results are "statistically significant"
Common values:
Default:

Statistical Significance

If p-value is small enough, it's very unlikely to get the results we observed if H_0 is true due to sampling variability
Results are statistically significant if they provide convincing evidence against H_0

Formal Statistical Decisions

- p-value ____ α: Reject H_0
- p-value ____ α: Do not reject H_0

Summary

1. Formulate hypotheses in terms of *population parameter*
2. Collect data and compute a *sample statistic*
3. Use the *sample statistic* to make a claim about the *hypotheses*

Today, we covered step 3:

3a. Construct *null/randomization distribution* using `permTest` in R or StatKey

If H_0 is __________________, what does the range of sample statistics look like?

3b. Compute *p-value*

Numeric summary of how extreme ____
is compared to ________________ distribution

3c. Make a formal decision about H_0

- Reject H_0 if ____________________________________
- Do not Reject H_0 ____________________________________

3d. Report your conclusion in context